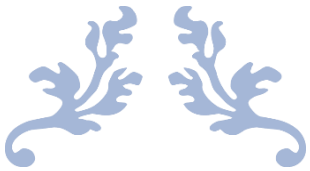

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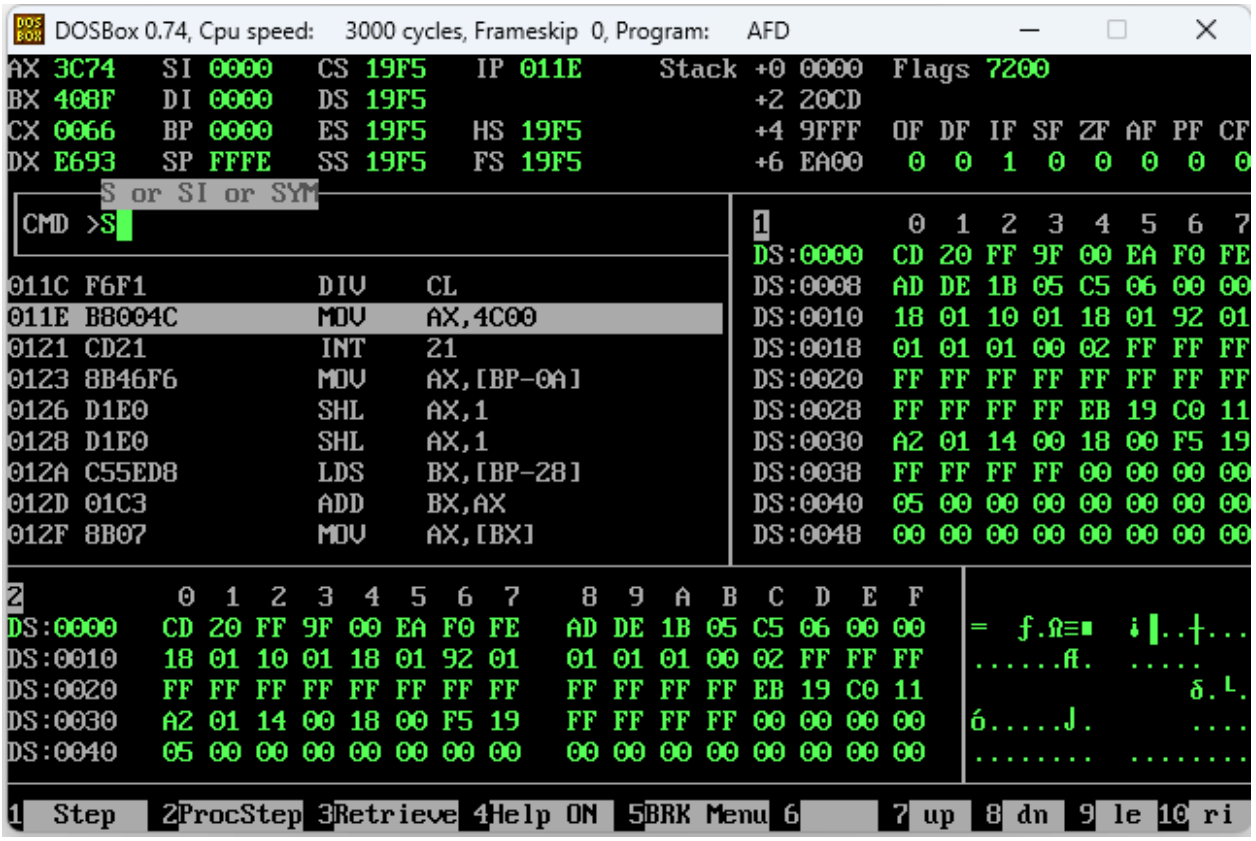
COAL LAB FINAL

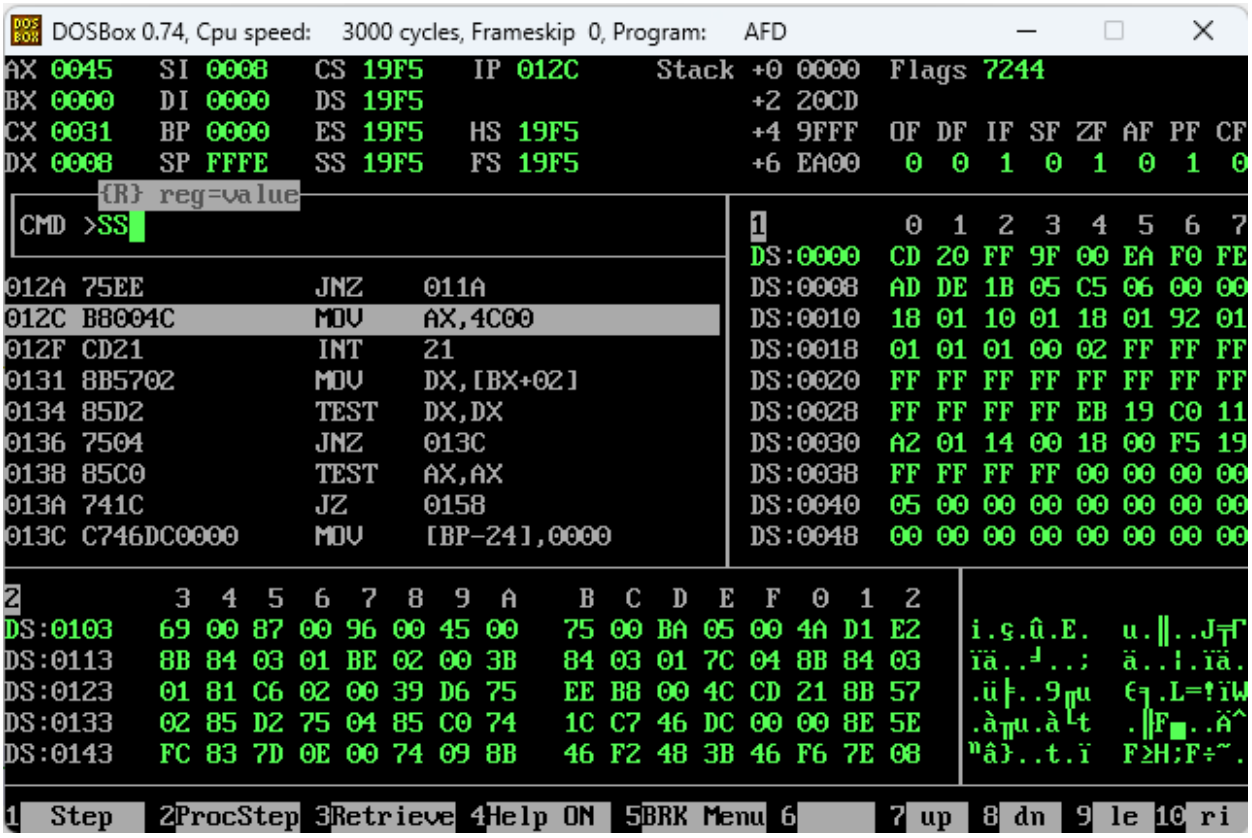
Q1(i) DRY RUN AND WRITE REGISTERS'S VALUE

```
org 0x100
jmp start

start:
        mov ax, 0xA193
        mov bx, 0x45AF
        mov cx, 0102
        mov dx, 0xE693

        xor ax,bx
        add ax,dx
        and bx,ax
        sub ah,dl
        mul  bl
        ror ax,3
        div cl


mov ax, 0x4c00
int 0x21
```



Q1(ii) Find the minimum element from array *69h,87h,96h,45h,75h,* don't use cx register.

```
org 0x100
jmp start
grade: dw 69h,87h,96h,45h,75h
start:

        mov dx, 5 ; length of array
        dec dx
        shl dx, 1           ;points to last index of array

        mov ax, [grade+si]
        mov si, 2
l1:
        cmp ax, [grade+si]
        jl skip
        mov ax, [grade+si]

skip:
        add si, 2
        cmp si, dx
        jne l1


mov ax, 0x4c00
int 0x21
```

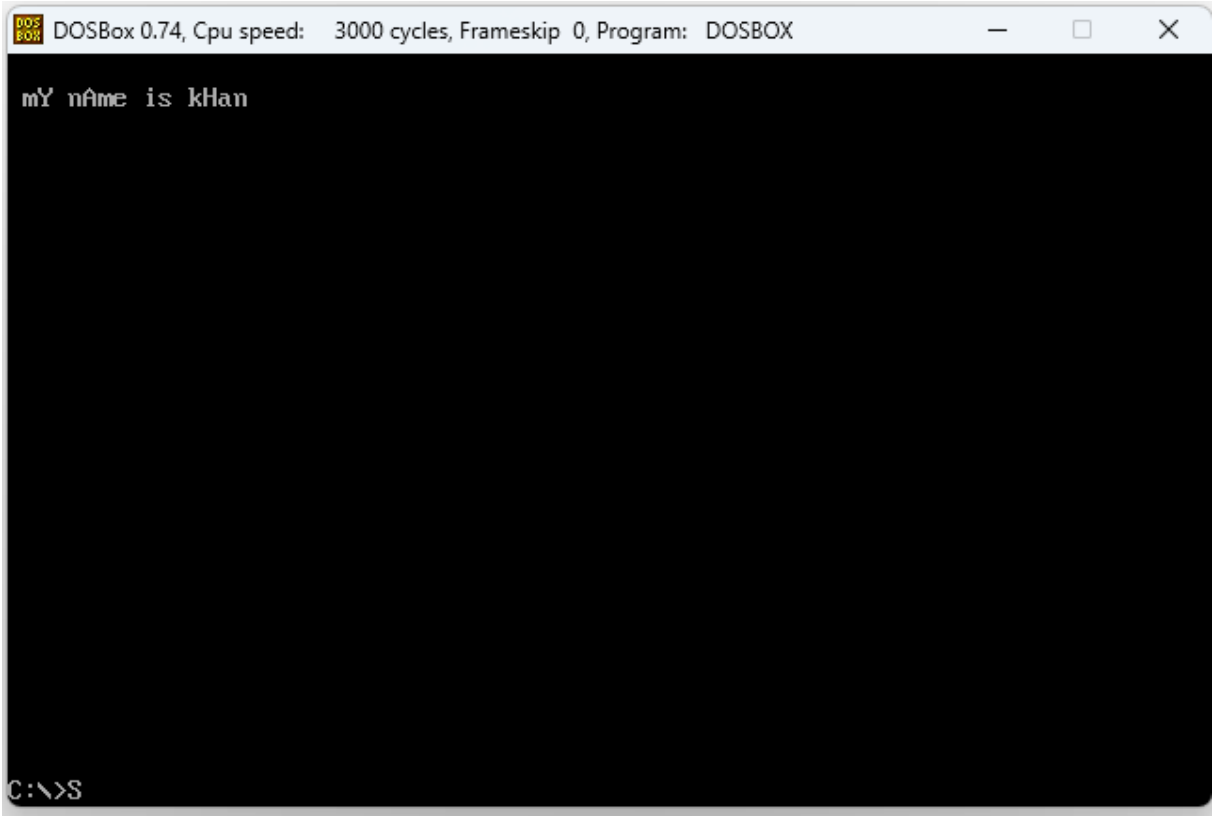


Ax has the lowest value, which is at 0109

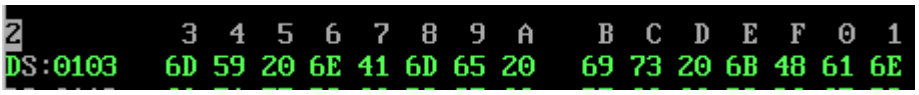
Q2 (i) Print the string mY nAme is kHan using BIOS Service. And convert the string to all capital letters.

```
org 0x100
jmp start
string: db 'mY nAme is kHan'
clearscreen:
        push ax
        push es
        push cx
        push di

        mov ax, 0xb800
        mov es, ax
        mov di,0

        mov cx, 2000
        mov ax, 0x0720

        rep stosw

        pop  di
        pop cx
        pop es
        pop ax
        ret

convert:
        push bp
        mov bp, sp
        push si
        push cx
        push ax

        mov  si,  [bp+6]
        mov cx, [bp+4]
        cld
l1:
        lodsb

        cmp al, 0x20
        je skip
        cmp al, 97
        jle skip
        add byte[si], 32
skip:

        loop l1
```

```
        pop  ax
        pop  cx
        pop   si
        pop bp
        ret 4
start:
call clearscreen

        mov ah, 0x13;service
        mov al, 1 ;sub service
        mov bh, 0 ; page
        mov bl, 07 ;attribute
        mov dx, 0x11;row col
        mov cx, 14 ;len

        push cs
        pop es

        mov bp, string
        int 0x10

;;;;;;

        mov ax, string  ;(+6)
        push ax
        mov ax, 15        ;length (+4)
        push ax
        call convert


mov ax, 0x4c00
int 0x21
```



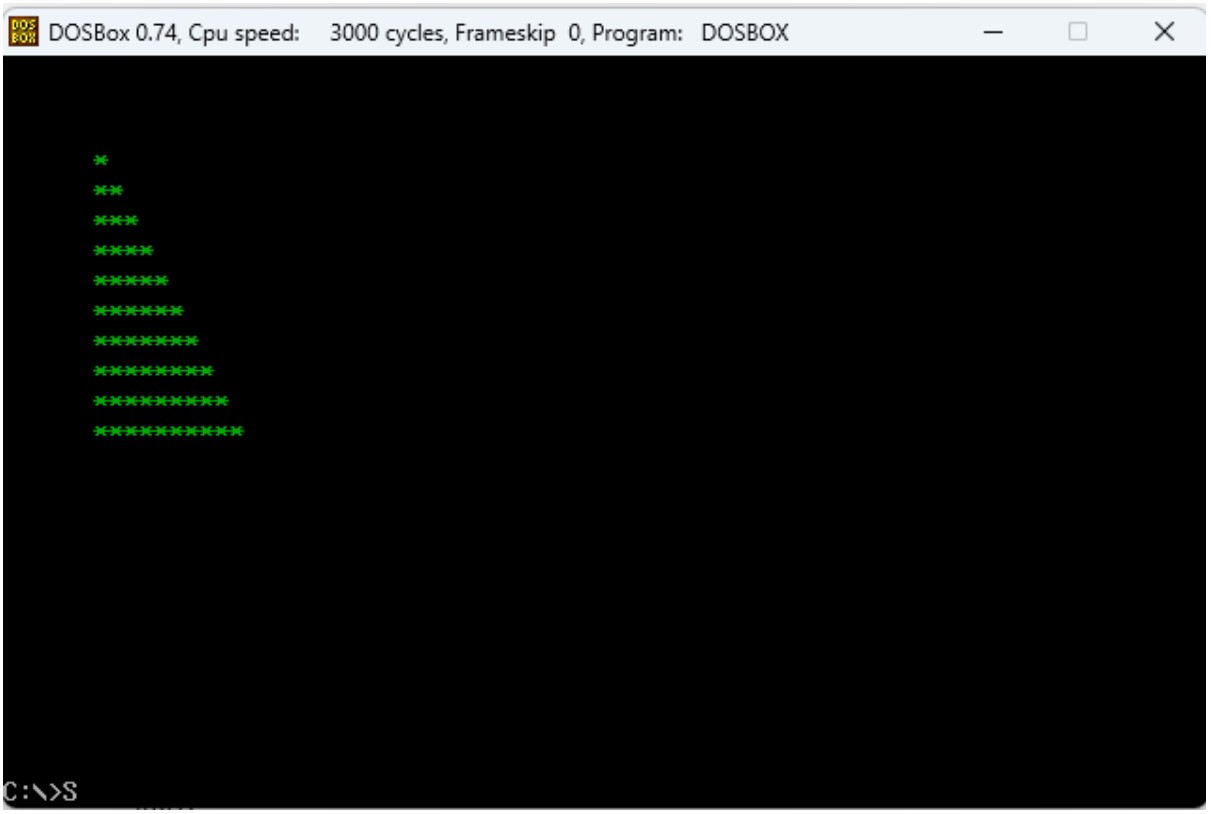
Askii of array



Q2 (ii) Print the triangle as given output, calculate location, clear screen, use subroutine and pass value through stack.

```
 org 0x100
 jmp start
 clearscreen:
        push ax
        push es
        push cx
        push di

        mov ax, 0xb800
        mov es, ax
        mov di,0

        mov cx, 2000
        mov ax, 0x0720

        rep stosw

        pop  di
        pop cx
        pop es
        pop ax
        ret

 triangle:
        push bp
        mov bp, sp
        push ax
        push cx
        push dx
        push es
        push di
        push si

        mov ax, 0xb800
        mov es, ax

        mov al, 80
        mul byte[bp+10]
        add ax, [bp+8]
        shl ax, 1
        mov di, ax        ;location
        mov si, ax  ;copy location

        mov cx, [bp+4]
        add cx, 1
        mov bx, 1         ; used to know the number of stars to print in on eline
        mov ax, [bp+6]

 nextline:
```

```
        mov di, si
        mov dx, 0

innerloop:
        mov [es:di], ax
        add di, 2
        inc dx
        cmp dx, bx
        jne innerloop

        add si, 160
        inc bx
        cmp bx, cx
        jl nextline


        pop   si
        pop  di
        pop  es
        pop  dx
        pop  cx
        pop  ax
        pop bp
        ret 8
start:
        call clearscreen

        mov ax, 4          ;y value +10
        push ax
        mov ax, 6          ;x value +8
        push ax
        mov ah, 0x02     ;attribute
        mov al, 0x2a      ;char
        push ax                    ;+6
        mov ax, 10 ;lenght        +4
        push ax

        call triangle

mov ax, 0x4c00
int 0x21
```


DOSBox 0.74, Cpu speed: 3000 cycles, Frameskip 0, Program: DOSBOX
*
**
***
****
*****
******
****
****
****
****
C:\>S

Q3 Use hardware interrupt, hook and print the string *fast cfd campus* when user press tab and when space is pressed print the reversed string, and terminate the program when ESC is pressed. Scan code of Tab (0x0f) and Space (0x39).

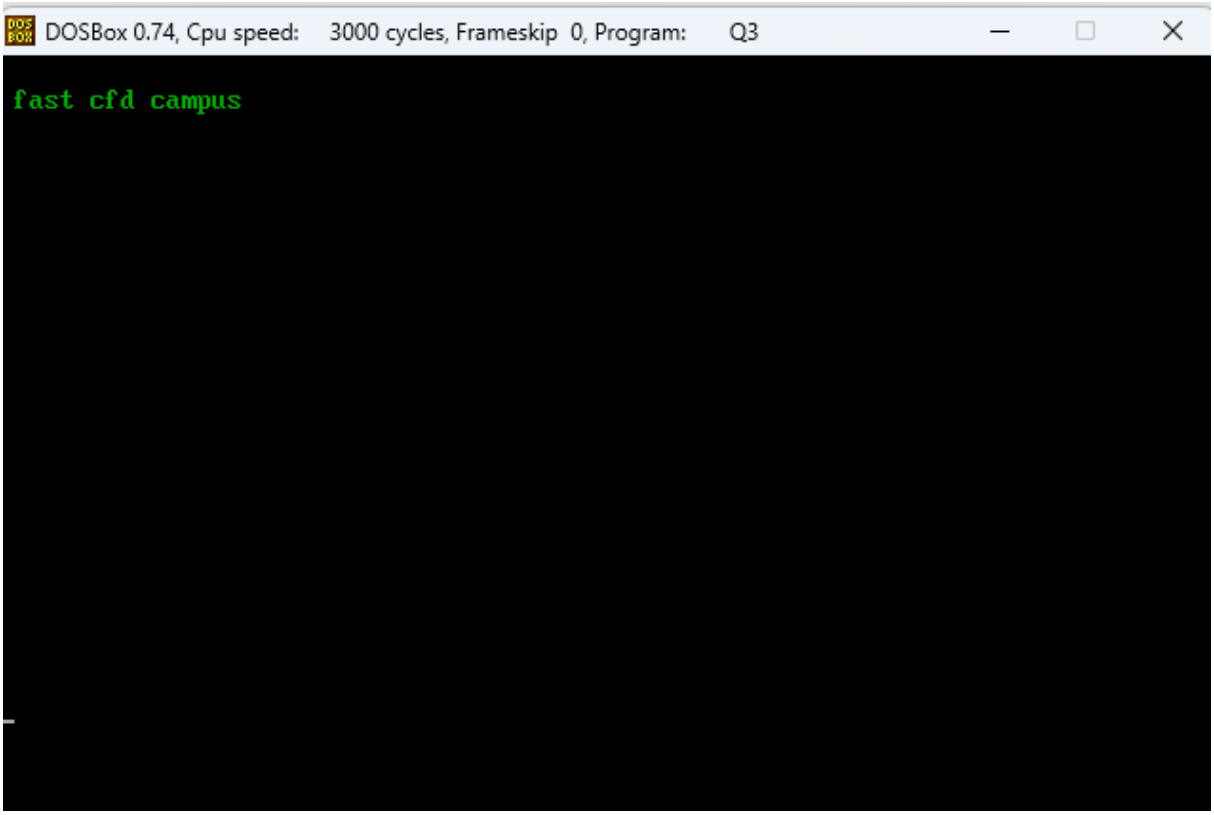
```
org 0x100
jmp start
string: db 'fast cfd campus', 0
oldisr: dd 0
clearscreen:
        push ax
        push es
        push cx
        push di

        mov ax, 0xb800
        mov es, ax
        mov di,0

        mov cx, 2000
        mov ax, 0x0720

        rep stosw

        pop  di
        pop cx
        pop es
        pop ax
        ret
;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;
printstr:
        push bp
        mov bp, sp
        push ax
        push cx
        push es
        push di
        push si

        mov ax, 0xb800
        mov es, ax

        mov al, 80
        mul byte[bp+8]
        add ax, [bp+6]
        shl ax, 1
        mov di, ax          ;location

        mov cx, [bp+12];length
        mov si, [bp+4]  ;offset
        mov ax, [bp+10]             ;attribute
        mov al, 0

        cld
```

```
next:
		lodsb
		stosw
		loop next

		pop   si
		pop   di
		pop  es
		pop  cx
		pop  ax
		pop bp
		ret 10
;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;
printrevstr:
		push bp
		mov bp, sp
		push ax
		push cx
		push es
		push di
		push si

		mov ax, 0xb800
		mov es, ax

		mov al, 80
		mul byte[bp+8]
		add ax, [bp+6]
		shl ax, 1
		mov di, ax		;location

		mov cx, [bp+12]		;length
		mov si, [bp+4]	;offset
		add si, 14 ;points to last of string
		mov ax, [bp+10]		;attribute
		mov al, 0


next1:
		std
		lodsb
		cld
		stosw
		loop next1

		pop   si
		pop   di
		pop  es
		pop  cx
		pop  ax
		pop bp
```

```
        ret 10

;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;

myKBISR:
        push ax
        push bx

        in al, 0x60         ;keystroke

        cmp al, 0x0f        ;tab
        jne nextkeycheck

;;;;;;
        mov bx, 15          ;length (+12)
        push bx
        mov bh, 0x02        ;attribute(+10)
        push bx
        mov bx, 1           ;y value (+8)
        push bx
        mov bx, 1           ;x value (+6)
        push bx
        mov bx, string ;(+4)
        push bx
        call printstr
;;;;;;

        jmp exit

nextkeycheck:
        cmp al, 0x39        ;space
        jne nomatch

;;;;;;
        mov bx, 15          ;length (+12)
        push bx
        mov bh, 0x02        ;attribute(+10)
        push bx
        mov bx, 1           ;y value (+8)
        push bx
        mov bx, 1           ;x value (+6)
        push bx
        mov bx, string ;(+4)
        push bx
        call printrevstr
;;;;;;

exit:
nomatch:

        pop bx
```

```
            pop ax
            jmp far [cs:oldisr]


;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;
start:
            call clearscreen

            mov ax, 0
            mov es, ax

            mov ax, [es:9*4]
            mov [oldisr], ax
            mov ax, [es:9*4+2]
            mov [oldisr+2], ax

            cli
            mov word [es:9*4], myKBISR
            mov word [es:9*4+2], cs
            sti

l1:
            mov ah, 0
            int 0x16

            cmp al, 27                      ;exit if esc is pressed
            jne l1

            mov ax, [oldisr]
            mov bx, [oldisr+2]

            cli
            mov word [es:9*4], ax
            mov word [es:9*4+2], bx
            sti

            mov ax, 0x4c00
            int 0x21
```

WHEN TAB IS PRESSED PRINTS STRING



WHEN SPACE IS PRESSED PRINTS REVERSE STRING